

$\text{sort_list_intersec}(LL, LS) :- \text{intersec}(LL, LR), \text{qsort}(LR, LS).$

Exemplu: $\text{sort_list_intersec}([[-4, -2, 3], [4, 7, -2], [-4, 7, 8, 4], [0, 1, 3, 7, 8], [-1, 7, 4, 9]], \text{Liste_resultat}).$
 $\text{Liste_resultat} = [1, 7]$

2. Sortarea listei rezultat după concatenarea a $n \in \mathbb{N}$ liste de valori reale.

$LL = [L_1, L_2, \dots, L_n]$, $L_i = [H_1^i, H_2^i, \dots, H_{n_i}^i]$, $i = 1 \dots n$, $n_i = \text{mărimea listei } i$.

$\text{Concaten}(LL) = \text{Concaten}([L_1, L_2, \dots, L_n]) = LR \xrightarrow{\text{sortare}} \text{LRS}$

Program Prolog

domains

lista = real *

lista_lista = lista *

predicates

concaten(lista, lista, lista)

concatenH(lista_lista, lista)

bsort(lista, lista)

schimba(lista, lista)

sort_concaten(lista_lista, lista)

clauses

$\text{concaten}([], L, L).$

$\text{concaten}([H|T], L2, [H|TR]) :- \text{concaten}(T, L2, TR).$

$\text{concatenH}([], []).$

$\text{concatenH}([HL|TL], LR) :- \text{concatenH}(TL, RTL), \text{sortare}(HL, RTL, LR).$

$\text{bsort}(L, LS) :- \text{schimba}(L, L1), \text{bsort}(L1, LS), !, \text{bsort}(L, L).$

$\text{schimba}([X, Y | T], [Y, X | T]) :- X > Y.$

$\text{schimba}([X | T], [X | T1]) :- \text{schimba}(T, T1).$

$\text{sort_concatenare}(LL, LRS) :- \text{concatenareH}(LL, LR), \text{bsort}(LR, LRS).$

Exemplu: $\text{sort_concatenare}([[1, -1, 7, 3], [2, 0, 4], [1, 10, 11], [7, 2]], \text{Liste_nr_sortata})$.